

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 4th Year 1st Term Regular Examination, 2019

URP 4111

(Utility Services and Infrastructure Planning)



Full Marks: 210

Time: 3 hrs

- N.B.**
- i) Answer any three questions from each section in separate script.
 - ii) Figures in the right margin indicate full marks.
 - iii) Assume reasonable value for any missing data.

Section – A

1. (a) Infrastructure should be described through a set of characteristics instead of a single definition. Explain the statement. (12)
(b) Write down the demographic factors that can influence the infrastructure decision. (10)
(c) What are the influential factors behind the infrastructure investment decision? Explain the role of the Sustainable Development Goals (SDGs) in determining and prioritizing investment decisions with relevant data and examples. (13)
2. (a) Explain the following terms with examples in connection with infrastructure and market failure: (15)
(i) Spillover benefits (ii) Negative externalities (iii) Monopoly
(b) Differentiate between infrastructure financing and funding with examples. (10)
(c) Explain the concept of BDO, BOO, BOT and concession with examples (10)
3. (a) Why is it important to set up guidelines for infrastructure service delivery? Write down the differences between output and input based guidelines. (10)
(b) How do we measure welfare of a community? What is Pareto Improvement and how is the concept of Pareto Improvement related with the economic evaluation of infrastructure project? (15)
(c) Explain how infrastructure investment can crowd in and crowd out private investment. (10)
4. Khulna city corporation (KCC) has received a development fund from GoB to construct a park for the city dwellers. Based on potential number of users, construction and operation cost there are four available alternatives to KCC for construction. The alternatives are: Children Park, Urban park, Linear Park and Theme park. Pair wise comparison of the mentioned criteria and the Ranking alternatives are given in the following tables: (35)

Criteria	Number of Users	Construction Cost	Maintenance Cost
Number of Users	1	1/2	3
Construction cost	2/1	1/1	4/1
Maintenance Cost	1/3	1/4	1/1

Number of Users				
	Children Park	Urban Park	Linear Park	Theme Park
Children Park	1/1	1/4	4/1	1/6
Urban Park	4/1	1/1	4/1	1/4
Linear Park	1/4	1/4	1/1	1/5
Theme Park	6/1	4/1	5/1	1/1

Maintenance Cost				
	Children Park	Urban Park	Linear Park	Theme Park
Children Park	1/1	2/1	5/1	1/1
Urban Park	1/2	1/1	3/1	2/1
Linear Park	1/5	1/3	1/1	1/4
Theme Park	1/1	1/2	4/1	1/1

Construction Cost	Cost in BDT (in millon)
Theme Park	34
Children Park	28
Urban Park	27
Linear Park	24

Select the best construction alternative for KCC using Analytical Hierarchy Process (AHP) method.

Section – B

5.

(a)

Differentiate between collective and residential utility services with example.

(05)

(b)

Enumerate the general considerations for planning and designing of Water Supply System.

(10)

(c)

Describe some Water Demand Management’s technical and policy measures that will reduce water crisis in Bangladesh. Provide necessary examples.

(15)

(d)

Write down the advantages and disadvantages of Gravitational water transmission and distribution system.

(05)
6.

(a)

Mention the locational criteria of an intake design.

(05)

(b)

Describe different principles and policies undertaken in Bangladesh to achieve sustainable water and sanitation system.

(10)

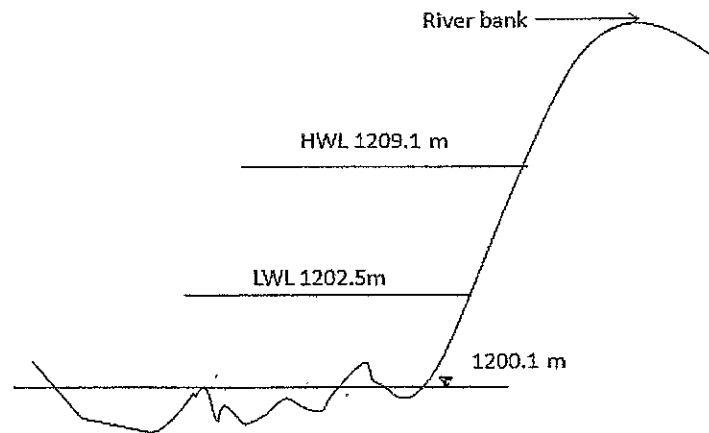
(c)

The specific water demands are estimated for region A, B and C excluding leakage losses. Estimate the demand in 15 years for each three of the regions considering leakage losses (given in the table below). Assume the annual demand growths are 5%, 3.5% and 1.5% respectively for A, B and C. Compare the results among the regions by applying the exponential model.

(20)

Regions	Population (age cohort) in years				Est. avg. water demand (l/c/d)	Leakage losses (of the est. avg. water demand)
	0-15	16-40	41-65	above 65		
A	5000	16377	7660	9753	180	5%
B	10000	3254	5243	8965	150	3%
C	8463	9000	5467	9000	176	4%

7. (a) Estimate the average daily water demand for a town having 110000 population. Assume industrial use 10%, institutional and commercial use 15%, public use 5% and livestock 10% of domestic demand. Take per capita consumption of 50 l/day and leakage 5%. (12)
- (b) Mention different community managed water supply systems of Bangladesh. (10)
- (c) Show various layout patterns of water distribution system. (08)
- (d) Write down the advantages and disadvantages of "pumping with storage" water transmission and distribution system. (05)
8. (a) What are the basic rules of fire fighting demand calculation and future water demand forecasting? (10)
- (b) How coagulation works in water supply process? Describe a full water supply system with diagram. (10)
- (c) Describe the pier intake system. Given the following information propose a suitable river intake. (15)
- Daily demand: 5000 m³
 - Pump capacity: 50 l/s (working 8 hours/day)



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BURP 4th Year 1st Term Regular Examination, 2019

URP 4161

(Project Planning and Management)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

- | | | |
|----|--|----|
| 1. | (a) Elaborate the followings (Any three): | 15 |
| | (i) Vision (ii) Mission (iii) Policy (iv) Plan (v) Strategy (vi) Programme | |
| | (b) How do projects are categorized in Bangladesh? Describe with necessary example. | 10 |
| | (c) Discuss the structure and functions of the planning commission in Bangladesh. | 10 |
| 2. | (a) Distinguish between private and public project with necessary examples. | 10 |
| | (b) Draw your concept on project life cycle. | 10 |
| | (c) Illustrate a hypothetical project fitting with various phases of project management cycle. | 15 |
| 3. | (a) Distinguish between project Monitoring and Evaluation | 10 |
| | (b) Discuss the importance of monitoring and evaluation with necessary example. | 10 |
| | (c) Discuss various types of evaluation of project with necessary example. | 15 |
| 4. | (a) Draw your concept on logical framework approach (LF) of project design. | 08 |
| | (b) Describe three important steps of Logical Framework approach of project design. | 15 |
| | (c) Describe various steps of Stakeholder Analysis of a development Project. | 12 |

Section – B

- | | | |
|----|--|----|
| 5. | (a) Mention different types of cash flow with necessary example. | 10 |
| | (b) Discuss the advantages and disadvantages of different investment criteria. | 15 |
| | (c) What is FVIFA? Discuss the application of future value annuity formula in our day to day life. | 10 |
| 6. | (a) What is an annuity? What are the differences between an ordinary annuity and an annuity due? | 08 |

- (b) Obtain the formula for future value of an annuity. 10
- (c) Phoenix Company borrows tk. 500,000 at an interest rate of 14%. The loan is to be repaid in 4 equal annual instalments payable at the end of each of the next 4 years. Prepare the loan amortization schedule. 10
- (d) Write a short note on Work Breakdown Structure (WBS). 07
7. (a) Mention the properties and limitations of NPV rule in short. 13
- (b) Consider the following cash flows of a project being considered by Techtron Limited. Now, find the IRR of this project when the Cost of Capital is 18%. 12

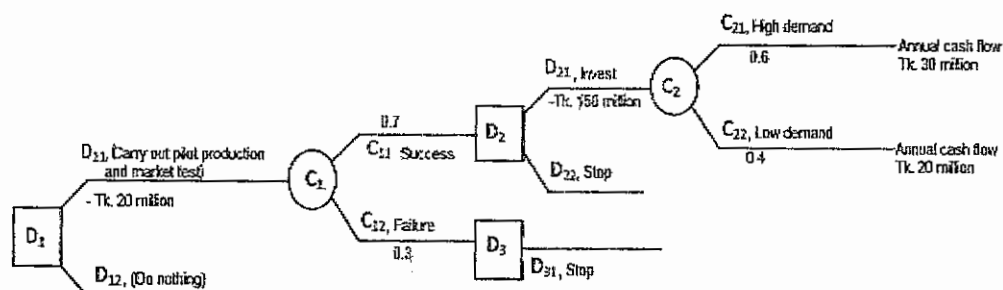
Year	0	1	2	3	4	5
Cash Flow	100,000	30,000	30,000	40,000	45,000	40,000

- (c) Illustrate the pros and cons of Payback Period in brief. 10
8. (a) Mention the differences between Equity and Debt. Illustrate the key factors in determining the Debt-Equity Ratio. 13
- (b) What is meant by Contribution Margin Ratio? Cash flow forecast for Naveen's Flour Mill Project is as follows: 12

	BDT in thousands	
	Year 0	Year 1 - 10
1. Investment	20,000	
2. Sales		18,000
3. Variable costs (66.67% of sales)		?
4. Fixed costs		1,000
5. Depreciations		2,000
6. Pre-tax profit		?
7. Taxes (30% of pre-tax profit)		?
8. Profit after tax		?
9. Cash flow from operation		?
10. Net cash flow		?

Now; complete the table above and calculate the Financial Break-even sales when the discount rate is 12%.

- (c) The following decisions tree illustrates an investment decision towards a new project. 10



Now, calculate the optimal course of actions using this tree. Assume, the discount rate is 12% and the project life is 20 years.

Note: Negative cash flow means cash outflow and positive cash flow means cash inflow.

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BURP 4th Year 1st Term Examination, 2019.
URP 4175
(Resettlement Planning)

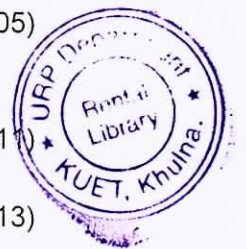
Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume data where necessary.

Section – A

1. (a) Define Displacement and Resettlement. What are the types of displacement? (10)
Explain in short.
(b) Why is resettlement planning an important issue for urban planners in a (10)
developing country like Bangladesh?
(c) What are the basic principles of Bangladesh constitution? How do you relate (15)
those components to resettlement planning?
2. (a) What is “3 Dhara” notice? Explain the major steps of land acquisition in (12)
Bangladesh (based on ARIPO, 1982).
(b) List out the possibility of corruption at every stage of the land acquisition (07)
process.
(c) Make a comparative analysis between the Acquisition and Requisition of (11)
Immovable Property Ordinance 1982 and Chittagong Hill Tracts Regulation
1958.
(d) Poor resettlement policy and lack of grievance redress mechanism led to (05)
arms struggle in the hill tracts region – explain.
3. (a) What are the key principles of resettlement? What are the major components (10)
of Resettlement Action Framework?
(b) Why is the WB Involuntary Resettlement Handbook a landmark policy (13)
document? Make a comparison between the WB resettlement policy and
ARIPO 1982.
(c) Why Jamuna Multi-purpose Bridge (JMB) is a landmark infrastructure project (11)
in Bangladesh? Mention the major limitations of the JMB resettlement policy.
4. (a) Suppose a person is to resettle and receive compensation. He/she owns 3 (15)
bigha agri-land with standing crop (rice), 0.5 bigha pond with fish, 0.2 bigha
of high land surrounding the pond, a house (pucca) on 0.25 bigha land, 0.5
bigha of mango-forest. Explain how the compensation will be estimated and
who will be responsible for the estimation. Also estimate the compensation
cost with assumed data.
(b) Bangladesh is striving for rapid industrialization, and infrastructure (10)
development is a way to facilitate industrialization. Explain how urbanization
and slummization is linked to resettlement and displacement.
(c) Explain the following terms: (10)
- Land based resettlement
- Wage based resettlement



Section – B

5. (a) What is resettlement? Why is there a need for resettlement and rehabilitation policy in Bangladesh? (15)
(b) Briefly describe the different types of resettlement. (10)
(c) Mention the powers of Deputy Commissioner and Arbitrators. (10)
6. (a) Why is property rights an issue in the Economics of Resettlement? (10)
(b) Explain the various loopholes in the land pricing process. (10)
(c) Describe the shortcomings of cost-benefit analysis in resettlement. (15)
7. (a) Write a case study showing difficulties and lessons learned during a large Involuntary Resettlement Project. (15)
(b) Write down the steps to undertake grievance redress mechanism. (10)
(c) Discuss the procedure for award of compensation by Deputy Commissioner. (10)
8. (a) Mention the penalties involved with land acquisition procedures. (07)
(b) Briefly explain the importance of IRR model in Resettlement. (10)
(c) Write short notes on: (10)
(i) Manipulation criteria in cost-benefit analysis.
(ii) Functions of IRR model
(d) "Grievance Redress Mechanism (GRM) design and implementation costs can vary significantly depending on the type, scale and scope of the GRM" – justify the statement. (08)

Khulna University of Engineering & Technology
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BURP 4th Year 1st Term Regular Examination, 2019

URP 4141
(Environmental Planning and Management)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Why the environment is considered as a system? Illustrate with necessary example. (10)
(b) Discuss various types and sources of pollutants. How these impact on the environment and society? (15)
(c) How do better environmental planning and management improve urban environmental quality? Discuss in brief. (10)
2. (a) Describe the historical evolution of the concept - Sustainable Development. (10)
(b) Illustrate the three pillars of sustainable development with necessary examples. (10)
(c) Discuss important principles and action areas of sustainable development. (15)
3. (a) What is the tragedy of common? Illustrate the characteristics of public goods. (10)
(b) Discuss environmental externalities with necessary example. How externalities could be internalized? Illustrate with example. (15)
(c) Illustrate the steps of environmental auditing. (10)
4. (a) Draw your concept on command and control instruments of environmental management. (10)
(b) Describe various market based instruments (MBI) of environmental management. (15)
(c) Discuss the role of various persuasive instruments of environmental management with necessary examples. (10)

Section – B

5. (a) Write down your understanding of political ecology with an example. (05)
(b) Describe in short the factors that should be considered to choose the EIA methods for environmental impact identification and prediction. (10)
(c) Explain the concept of normal variability and the current trend to describe the baseline situation of the environment in the context of carrying out EIA with an example. (10)
(d) "A development project with relation to EIA goes through different stages from project identification to environmental audit" - Show the relationship among the different stages by drawing a proper figure and explain it briefly. (10)
6. (a) Why is Environment a major concern in city areas than in rural areas? (05)
(b) Significance of an impact = [Environmental Value x Impact Magnitude]. Explain the equation to calculate the significance of an impact on a biophysical system or component of the environment while conducting EIA. (10)
(c) Explain in short the common mitigation measures of air pollution during plant operation under the Environmental Management Plan (EMP). (10)
(d) Briefly describe the guiding principles for Green Belt Development as a protective measure against the environmental impact under Environmental Management Plan (EMP). (10)
7. (a) Differentiate between model calibration and model validation with a connection to EIA impact prediction taking the mathematical modeling approach as an example. (10)
(b) Predicted impact magnitude in most cases follows an objective approach and expresses the explicit unit where the determination of significance follows the subjective approach. Do you agree with this statement? Provide arguments in favor of your answer by an example. (10)

- (c) A completely mixed lake receives two inflows: natural stream flow $0.2 \text{ m}^3/\text{s}$, wastewater discharge $0.074 \text{ m}^3/\text{s}$ and has a constant volume of $3 \times 10^6 \text{ m}^3$. Provided that the wastewater has $15 \text{ mg/l NH}_3\text{-N}$ and stream has $3 \text{ mg/l NH}_3\text{-N}$. (15)



Bacteria in the lake convert NH_3 to NO_3^- by a process called nitrification. $-r_N = k \cdot \text{CN}$, where k = a first-order rate constant = 0.03 day^{-1} and CN = concentration of ammonia-nitrogen mg/l . Find the lake and outflow $\text{NH}_3\text{-N}$. Ammonia is very toxic to fish and it is $1.5 \text{ mg/l NH}_3\text{-N}$. Does the amount of natural nitrification in the lake allow wastewater discharge of $25 \text{ mg/l Ammonia-N}$?

8. (a) Write down the advantages and disadvantages of overlay and network methods of impact identification. (10)
- (b) What are the limitations of a quantified matrix? How can a weighted matrix be used to eliminate the limitations of a quantified matrix? Explain your answer with an example. (10)
- (c) In an EIA project, primary, secondary and subsequent impacts (initial condition, consequent condition, and their effects) are identified. Total Impacts are classified into three categories viz. Branch 1, 2 and 3. Actions and Impacts of three branches along with their magnitude and importance with a probability of occurrence are also determined. Necessary data are provided in Table B₁, Table B₂, and Table B₃. (15)
- Using Network method of impact identification, calculate the network index for each branch along with the grand index. Compare and provide your insights on the results of the impacts on three branches.

Table B₁: Possible adverse impacts

Initial Condition	Consequent Condition	Effects
Increased surface runoff (E)	Flooding (H)	Gullyng and Erosion (K)
Pollution of Ground Water (F)	Degradation of Water Supply (I)	Health Hazard (L)
Removal of Topsoil (G)	Decreased Fertility (J)	Death of Flora (M)

Table B₂: Relation between action and impacts of three branches

Action	Impacts	Branches
A	(F) → (I) → (L)	1
B	(E) → (H) → (K)	2
C		
D	(G) → (J) → (M)	3

Table B₃: Relation among different type of impacts, their magnitude and importance with the probability of occurrence

Impacts	Impact Score (1-10 interval Scale)		Probability of Occurrence			
	Magnitude	Importance				
E	5	3	B → E(0.8)	D → E(0.7)		
F	2	5	A → F(.5)	B → F(0.4)		
G	3	4	C → G(.3)	D → G(.4)		
H	4	5	E → H(.7)			
I	2	9	F → I(.6)			
J	2	5	G → J(.8)			
K	3	7	H → K(.7)			
L	2	10	I → L(.9)			
M	1	6	J → M(.8)			

Khulna University of Engineering & Technology
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BURP 4th Year 1st Term Examination, 2019
URP 4171
(Legal Basis of Planning)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Elaborate the significance of legal aspects in the field of planning for a country like Bangladesh. (08)
(b) Make a comparative analysis between "Major Acts" and "Minor Acts" in Bangladesh. Planning laws are listed within which category mentioned above? (10)
(c) "Categorization of zones and blocks are more detailed and planning regulations are adopted according to the need of development project". Explain this statement with appropriate example. (10)
(d) Illustrate the chronological statute formation process on local government in India. (07)
2. (a) What is your understanding about "Eminent Domain"? Why is land acquisition termed as "Eminent Domain"? (05)
(b) Make a comparison between "Normal Acquisition" and "Emergency Acquisition" addressing the process of formation and execution. (12)
(c) How will you explain regulatory intervention in land acquisition practice? Provide appropriate example. (10)
(d) Write about the penalties and associated problems of land acquisition. (08)
3. (a) "Private Housing Project Land Development Rule, 2004" is applicable for which sectors of our country? Write about the stakeholders of the rule. (06)
(b) Specify and elaborate the issues addressed in the consultancy phase of Private Housing Project. (06)
(c) Make a list of clearance certificates necessary to submit for the approval of private housing project. (08)
(d) Critically analyze the significance and shortcomings of "Mega City, Divisional Town and District Town's municipal areas including country's all the municipal areas' Playground, Open Space, Park and Natural Water Reservoir Conservation Act, 2000 ". (15)
4. (a) Demarcate the role of following stakeholders in the formation of Locally-led New Town Development Corporation (LLNTDC) for designating New Town: (15)
i) Local Authority ii) Secretary of State iii) Local Community.
(b) Write short notes on the following issues of designating a New Town according to New Town Act, 1981 (Local Authority Oversight) Regulations 2018 (any two): (20)
i) Stewardship and community involvement for sustainable new town development
ii) Considerations for the appointment of Board Members (LLNTDC)
iii) Groups associated with Government consultation
iv) Documents necessary to attach with the proposal for designating new town and appointing oversight authority.

Section – B

5. (a) What is Bangladesh National Building Code (BNBC)? (05)
(b) Briefly discuss the planning aspects that have been considered in updated BNBC 1993. (20)
(c) Formulation of BNBC is a huge task of multi-professionals and experts. Evaluate this statement with examples. (10)
6. (a) As an Urban Planner critically analyze the causes, consequences and remedies of continuous deviation of building constructions in the urban areas of Bangladesh. (15)
(b) Discuss the governance and functions of RAJUK. (10)
(c) Narrate the importance of Planning Inspectors in comparison with Building Inspectors in Bangladesh. (10)
7. (a) Preparation and implementation of Master Plan and Site Development Schemes are the pre-requisites for planned development of a municipality. Evaluate in terms of regulatory provisions of the Pourashava Ordinance 2010. (15)
(b) Write notes on the followings: (20)
 - i) Sources of fund for a municipality
 - ii) Regulation of buildings under the Pourashava Ordinance 2010.
8. (a) Formulation and execution of URP Act 2017 requires coordinated efforts of many interrelated ministries and departments of Bangladesh. Critically evaluate this statement. (15)
(b) Narrate the functions of the Urban and Regional Planning Executive Council under the draft URP Act 2017. (10)
(c) Discuss the duties and responsibilities of Planning Inspectors under the URP Act 2011 of Zambia. (10)